- No mode, if all values occur equally often

* Comparison of Mean, Median, and Mode

Mean : uses all values and is affected by extreme values.

Median : represents the middle value and is less affected by extreme values.

Mode : identifies the most frequently occurring values.

* Example of Salary Analysis.

Suppose five people have salaries of 20,000, 25000, 30,000, 35,000 and 100,000.

The mean is 42,000.
The median is 30,000.
The mean is higher because the salary of 100,000 increases the average.

Therefore, the median can sometimes represent typical income better than the mean.

Conclusion: Mean, median, and mode are useful for understanding the center of a dataset.

~~Measure of Dispres~~

Measures of Dispersion

Measures of dispersion describe how spread out the values in a dataset are.

A. Range: The difference between the maximum and minimum values.

Formula: Range = Maximum value - Minimum value.

B. Variance: Measures the average squared deviation of values from the mean.

Population variance: $\sigma^2 = \Sigma(x-\mu)^2 / N$

Sample variance: $s^2 = \Sigma(x-\bar{x})^2 / (n-1)$

C. Standard Deviation : Measures how much data values vary around the mean.

formula : Standard Deviation = √Variance

D. Interquartile Range (IQR) : Measure the spread of the middle 50% of data.

formula : IQR = Q3 - Q1

2. QUARTILES AND PERCENTILES

Quartiles : Divide an ordered dataset into four parts.

- Q1 : First quartile (25th percentile).
- Q2 : Second quartile or median (50th percentile).
- Q3 : Third quartile (75th percentile).

Percentiles : Divide an ordered dataset into 100 parts and show the relative position of a value.

3. OUTLIERS

An outlier is a value that is unusually far from other observations.

Lower Bound $= Q_1 - 1.5 \times IQR$

Upper Bound $= Q3 + 1.5 \times IQR$

values below the lower bound or above the upper bound are potential outliers.

4. SKEWNESS

Skewness describes the asymmetry of a data distribution.

- Symmetric : Both sides are approximately balanced.
- Left-skewed : The distribution has a longer tail on the left.
- Right-skewed : The distribution has longer tail on the right.

Covariance, Correlation and Normal Distribution.

* Covariance

Covariance measures how two variables vary together.

- Positive covariance: variable tend to increase together.
- Negative covariance: one variable tends to increase as the other decreases.
- Zero covariance: Nolinear co-variation is indicated.

* Correlation

Correlation measures the strength and direction of the linear relationship between two variables.

The correlation coefficient is represented by r.

Its values ranges from −1 to +1.

- $r = +1$: Perfect positive correlation
- $r = -1$: Perfect Negative correlation.
- $r = 0$: No linear correlation.

Important : correlation does not necessarily imply causation.

✱ Z-Score

A z-score indicates how many standard deviations a value is above or below the mean.

Formula :

$$z = (x - \mu) / \sigma$$

where :

- x : observed value
- μ = mean
- σ = standard deviation

Interpretation:

- Positive Z-score: value is above the mean
- Negative Z-score: value is below the mean
- Zero Z-score: value equal the mean.

8. Normal Distribution

A normal distribution is a continuous, symmetric, bell-shaped probability distribution.

properties:

1. It is bell-shaped and symmetric.
2. Mean = Median = Mode.
3. Total area under the curve equal 1.
4. Approximately 68% of values lie within 1 standard deviation of the mean.
5. Approximately 95% lie within 2 standard deviations.
6. Approximately 99.7% lie within 3 standard deviations.

This is known as the 68-95-99.7 Rule.

9. Statistical Inference

Statistical inference uses sample data to draw conclusions about a population.

A. Hypothesis Testing: A method used to test a claim about a population.

- Hypothesis Testing: A method used to test a claim about a population.

[redacted]

- Null Hypothesis (H_0): usually represents no effect or no difference.
- Alternative Hypothesis (H_1): Represents an effect or difference.

B. P-value: Measures how compatible the observed data are with the null hypothesis.

C. Confidence Interval: A range of value used to estimate a population parameter.

D. Central limit theorem (CLT): Under suitable conditions the sampling distribution of the sample mean becomes approximately normal as sample size increases.